

MPNet: A modular deep learning process TCAD surrogate modeling framework

Qipei Zhang¹, Pengwei Liu², Wenzhang Fang¹, Dong Ni¹, and Yuting Kong^{1,†}

¹College of Integrated Circuits, Zhejiang University, Hangzhou 311200, China

²College of Control Science and Engineering, Zhejiang University, Hangzhou 310027, China

Abstract: The computational cost of TCAD simulations is becoming prohibitively high with the complexity of advanced process technologies, making simulation acceleration a critical research priority. While end-to-end surrogate models mapping process recipes to device structures and characteristics offer a promising alternative, their application is often limited by poor generalizability and explainability. In this work, we present MPNet, a modular deep learning surrogate modeling framework for process TCAD. MPNet comprises distinct surrogate models for individual process modules, which are assembled into an integrated framework. These modular models employ a novel UNet-attention feature evolution method to capture the complex evolutions of device geometry and doping profiles. Each module can be trained separately on its individual process, after which the modules are cascaded and jointly fine-tuned to minimize error accumulation throughout the cascade. The efficacy of the proposed MPNet framework is demonstrated through a MOSFET integrated process TCAD case study. Results show that MPNet achieves a computational speedup of over 10^3 times compared to conventional TCAD, while maintaining predictive fidelity exceeding 98%. Finally, to illustrate the application of the proposed framework, MPNet is coupled with a PSO algorithm, showcasing its utility for fast process optimization to meet specific process targets.

Key words: modular surrogate model; deep learning; process TCAD; PSO

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1. Introduction

IC (integrated circuits) manufacturing is an intricate and large-scale engineering process that requires significant time and resource costs. Generally, engineers use TCAD (technology computer-aided design) to mitigate these costs and accelerate development. As an essential tool, TCAD allows engineers to simulate device physics and predict the outcomes of fabrication steps without processing actual wafers.

However, with the scaling down of the transistors, the increasing complexity of IC manufacturing processes leads to a significant increase in TCAD simulation time, which limits its application in industrial-scale manufacturing. As accurate simulation typically requires solving complex systems of PDEs (partial differential equations) on extremely fine-grained meshes, a single integrated process flow can take hours or even days to complete. This computational bottleneck renders numerical TCAD impractical for applications requiring rapid iteration and large-scale sampling, such as PWO (process window optimization), statistical variability analysis, or robust DTCO, which all demand thousands of simulation runs^[1, 2].

To address these problems, deep learning models have emerged as a key approach, with data-driven surrogate models becoming a promising alternative^[3–6]. These models learn the complex mapping between process recipes and process

results, providing near-instantaneous predictions once trained. Research on TCAD process surrogate models predominantly focuses on end-to-end models. These models typically attempt to either: (1) directly map process recipes to final device electrical characteristics, using CNN or GNN methods to map process parameters and device I - V curves and electric field distributions^[7–10], or (2) model a single, isolated process step, such as predicting final doping profiles or etching results^[7, 11–15]. For example, the classic ANN model is used to implement typical process parameters and device electrical test results, but its explainability is reduced due to the lack of sufficient complete input parameters and test information^[16]. 3D CNN demonstrates advantages in feature extraction of TCAD 3D devices, which significantly reduces the number of Newton iterations^[17]. The DeePLabV3 model is proposed in MLTCAD, which predicts the average electric field of transistors. It accelerates the prediction of the FeFet (ferroelectric field-effect transistor) average electric field, which is 13 600 times faster than TCAD simulation^[7]. Samsung Electronics introduced RTT, a model that predicts the S/D doping profiles of the MOSFET using 2-D CNN and MLP^[18]. RNNs have also shown promise in modeling sequential fabrication steps by leveraging temporal dependencies in process data, whose framework can serve as inspiration for integrated process, but its application is still limited to the temporal changes of single-step processes and has not yet been extended to full integrated process flows^[19].

Despite their utility, the adoption of these end-to-end models for integrated process flows is hindered by significant limitations. Their 'black box' structure, which directly

Correspondence to: Y T Kong, ytkong@zju.edu.cn

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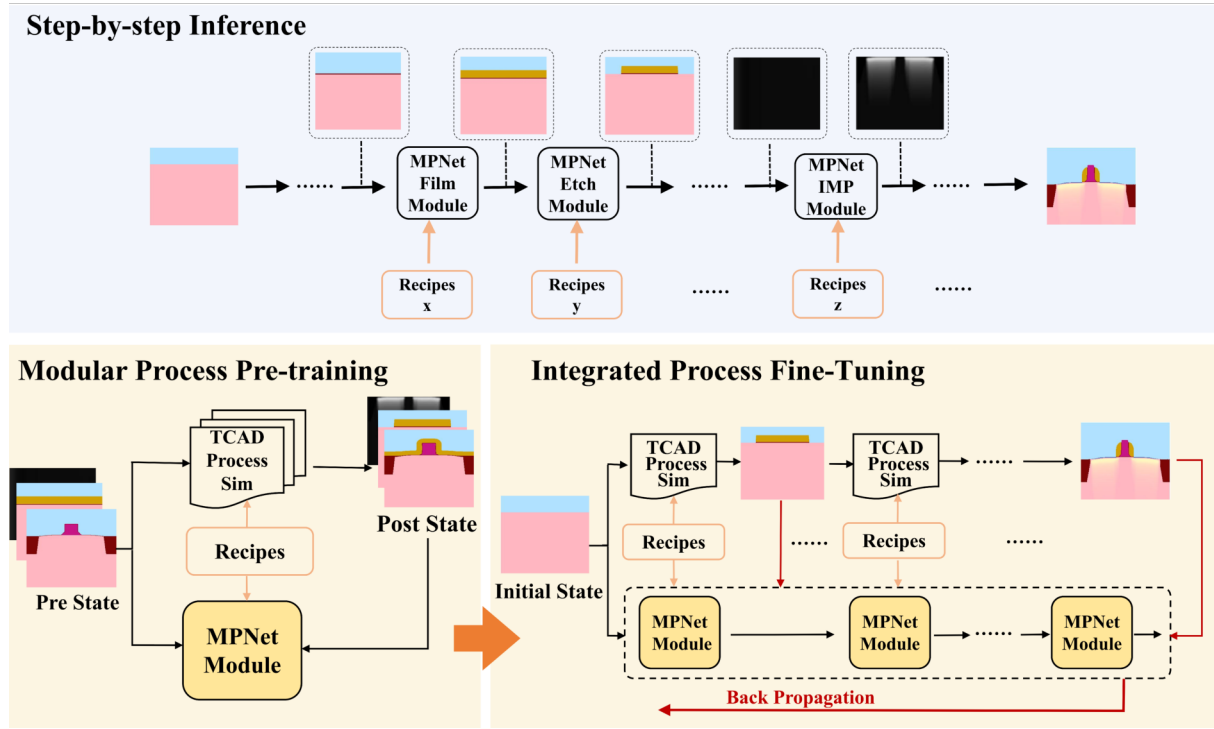


Fig. 1. (Color online) The framework of MPNet.

maps the input and output of the integrated process, inherently leads to poor flexibility and explainability. On one hand, by collapsing the entire multi-step sequence into a single prediction, these models obscure the critical step-by-step evolution of device geometry and doping profiles. On the other hand, the models lack process robustness: they must be completely retrained if intermediate steps are added or subtracted from the integrated process. This lack of physical transparency and validation of intermediate process steps makes it difficult to establish trust in the generalizability of the predictions for complex integrated fabrication steps.

In this study, we propose MPNet, the first modular deep learning process TCAD surrogate framework. Unlike end-to-end approaches, MPNet sequentially predicts process geometry and ion distribution throughout the integrated process flow. MPNet comprises surrogates for modular processes, each pre-trained to predict a specific process. These modules are cascaded and fine-tuned to perform as a surrogate for an integrated process. Our framework is demonstrated through a case study of an integrated MOSFET process (a traditional planar process). It achieves a considerable speed improvement while ensuring high accuracy, offering an explainable solution for rapid process adaptation.

The remainder of this paper is organized as follows. Section 2 introduces the dataset preparation method and provides detailed explanations of our MPNet framework's architecture with its training method. Section 3 presents comprehensive experiments to verify the outstanding performance of our framework. Section 4 validates the framework through a target process optimization case study. Finally, Section 5 concludes this paper.

2. Methodology

In this section, the methodology for the MPNet framework is detailed. First, a formal problem definition for the sur-

rogate modeling task is presented. The overarching MPNet framework, centered on its core modular pre-training and integrated fine-tuning strategy, is then introduced. Following this, the architecture's core components are detailed: the process surrogate module mechanism is described first, followed by an explanation of how these modules are cascaded and jointly fine-tuned into the integrated process surrogate model to mitigate cumulative error. Finally, the dataset preparing methods used for training are outlined.

2.1. Problem definition

This study addresses the critical need for a surrogate modeling framework that balances the competing demands of predictive accuracy and physical explainability in the integrated TCAD process. The central problem is to design and implement a modular deep learning architecture where each module surrogates a distinct TCAD process step. This necessitates a framework capable of sequentially predicting the step-by-step evolution of device geometry and doping profiles, while ensuring that the cascaded predictions remain stable and accurate across the entire integrated process flow.

Specifically, given a multi-step integrated MOSFET process consisting of N sequential steps.

- **Initial condition:** The process starts from a known initial process state X_0 (e.g., the bare substrate).

- **Inputs:** The model receives a sequence of process recipes $\mathbf{C} = \{C_1, C_2, \dots, C_N\}$, where each C_k is the set of parameters for step k (e.g., implant dose, etch time).

- **Ground truth:** The TCAD simulation produces a ground-truth sequence of process states $\mathbf{X} = \{X_1, X_2, \dots, X_N\}$, where each state X_k results from applying step k to the previous state X_{k-1} .

- **Surrogate Task:** We aim to construct a modular surrogate framework F composed of N corresponding modules, $F = \{\hat{m}_1, \hat{m}_2, \dots, \hat{m}_N\}$.

- **Outputs:** The framework F must take X_0 and $\mathbf{C}$ to sequen-

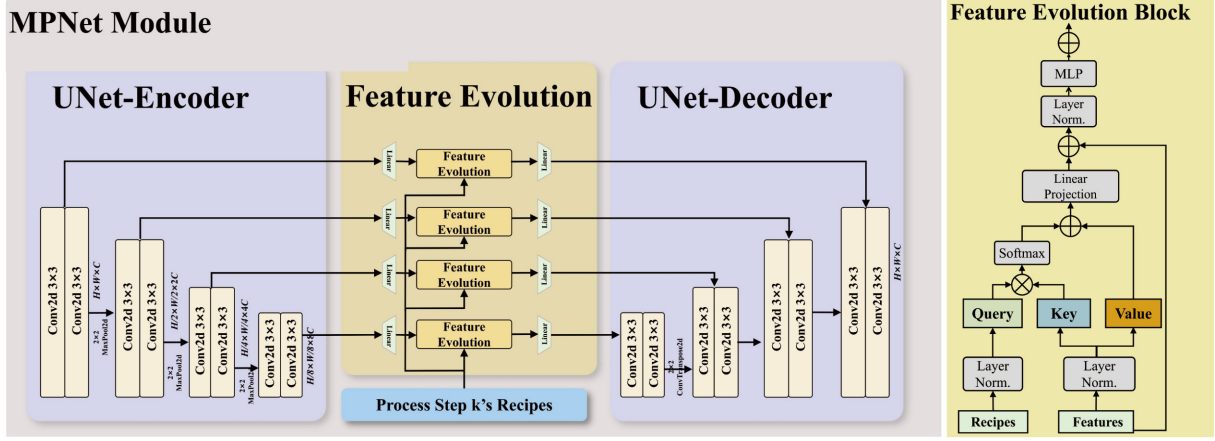


Fig. 2. (Color online) The module of MPNet, composed of a UNet and a cross-attention-based feature evolution block.

tially predict the integrated process of device states $\hat{X} = \{\hat{X}_1, \hat{X}_2, \dots, \hat{X}_N\}$, where each intermediate state is predicted as $\hat{X} = \hat{m}_k(\hat{X}_{k-1}, C_k)$.

2.2. MPNet framework

The workflow of our MPNet, as illustrated in Fig. 1, is based on a dual-phase training strategy designed to optimize both individual module accuracy and overall integrated process fidelity. This strategy consists of modular process pre-training followed by integrated process fine-tuning.

In the initial pre-training phase, individual MPNet modules are trained independently. Each module learns to accurately surrogate a single-step TCAD Process Sim by mapping a Pre State and its corresponding recipes to a ground-truth Post State. Following pre-training, these modules are cascaded in sequence to represent the integrated process flow. This integrated model then undergoes the fine-tuning phase, where back propagation is applied across all modules simultaneously. This critical joint optimization step adjusts parameters throughout the cascade to minimize the cumulative error. As depicted in the diagram, this architecture also supports parameter sharing, where modules performing similar process types utilize shared weights.

Once fully trained, the framework performs rapid step-by-step Inference. Starting from an initial state, the output of one module is sequentially fed as the input to the next, enabling the step-by-step prediction of the entire integrated process flow.

2.3. Modular process surrogate model

The core component of our framework is the MPNet modules designed for modeling single-step TCAD processes. As illustrated in Fig. 2, each MPNet unit utilizes a UNet architecture enhanced with a cross-attention-based feature evolution mechanism to integrate inputs and process recipes.

The UNet architecture serves as the foundational process translation framework for the process module within MPNet. Its structure is composed of a multi-layered encoder, which extracts hierarchical feature maps from the input NMOS device image, and a decoder, which reconstructs these features via deconvolution to produce the final output. Each MPNet unit employs a novel cross-attention-based feature evolution mechanism to integrate process recipes as a condition into the encoder's multi-level outputs. Specifically, this is implemented as a cross-attention block where the process recipes

are embedded to form the "Query", while the hierarchical features extracted by the UNet-encoder serve as the "Key" and "Value" pairs. This approach allows the model to effectively fuse multimodal data—the process condition recipes and the input image features—within the latent space. embedded process recipes are denoted by $\vec{C}$, the UNet's bottleneck output is $\vec{B}$, then we can rephrase it with the following equation:

$$\text{Attention}(\vec{C}, \vec{B}, \vec{B}) = \text{Softmax}\left(\frac{\vec{C}\vec{B}^T}{\sqrt{d_b}}\right)\vec{B}. \quad (1)$$

Cross attention helps the model to learn the intrinsic relationship between process recipes and input features, enabling feature selection aligned with process requirements.

This modular unit forms the building block of MPNet and is trained individually. The goal of this pre-training is to teach the module to accurately surrogate its specific process. During this pre-training phase, CE (cross-entropy loss) is used as the loss function of structure process models, and MSE (mean squared error) is used as the loss function of the implant process. Additionally, during feature extraction, to better capture changes in the processing region, we manually select the regions corresponding to each processing step, such as the bulk region for ion implantation and the gate region for polysilicon etching. During training, pixel loss in key regions is amplified to improve model performance in areas undergoing significant changes. The weighted loss function, L_{weighted} can be defined as:

$$L_{\text{weighted_CE}} = -\frac{1}{N} \sum_{i=1}^N W_i \cdot [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]. \quad (2)$$

or

$$L_{\text{weighted_MSE}} = \frac{1}{N} \sum_{i=1}^N W_i \cdot (\hat{y}_i - y_i)^2. \quad (3)$$

Here N is the total number of pixels, y_i is the ground truth label/value for pixel i , $\hat{y}_i$ is the predicted label/value, and W is a weight matrix. For a pixel i located in a key processing region, its corresponding weight W_i is set to k , which is the ratio of total input area to key area of this process step, while for pixels in non-key regions, $W_i = 1$. This ensures that the model pays more attention to the accuracy of critical areas.

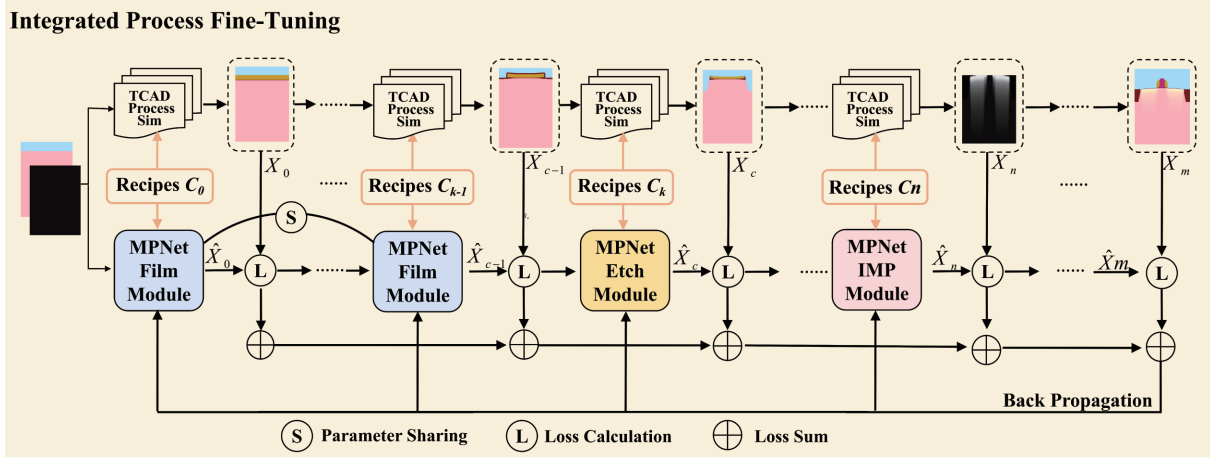


Fig. 3. (Color online) Fine-tuned MPNet for integrated process.

In TCAD simulation for semiconductor manufacturing, the same simulation function often performs similar process tasks. For example, dry etching machines can be used for silicon etching in the STI isolation region or for gate polysilicon etching, with the primary difference being the material being etched. Therefore, based on the process type in TCAD, whose TCAD simulation command corresponds to:

$$[\text{Process type}][\text{Species}][\text{Material}][\text{Recipes}]. \quad (4)$$

The pre-trained process models are categorized into etch, implant, and film modules. For modules of the same process type (e.g., all etch modules), the network parameters are shared, meaning they are identical and updated together during training. Specific etching materials are classified and embedded as process conditions for feature evolution, aligning with the actual operational scenario of the machine. This method can also reduce the parameters of the entire integrated process model.

Concurrently, in the FAB environment, it is rare to encounter processing steps that are entirely homogenous. Therefore, for processes of the same type but with significant practical differences, the modules within our MPNet can also be effectively decoupled. This decoupling can be implemented by amplifying the parametric divergence between distinct units, thereby facilitating a more reliable and robust training process.

2.4. Integrated process surrogate model

After pre-training the individual MPNet units, we construct the integrated process surrogate model by cascading the process modules according to the actual fabrication sequence, where the output of one model serves as the input for the next.

As shown in Fig. 3, let the integrated process consist of K steps, with the initial state being X_0 . The predicted state at step k , denoted as $\hat{X}_k$, is generated by the corresponding module M_k with recipes C_k :

$$\hat{X}_k = M_k(\hat{X}_{k-1}, C_k) \quad \text{for } k = 1, 2, \dots, K. \quad (5)$$

Here $\hat{X}_{k-1}$ is the predicted output from the previous step (with $\hat{X}_0 = X_0$). The final output of the integrated model is $\hat{X}_K$.

However, the input used for training the process mod-

ules is the output from the TCAD simulation, which does not include the errors potentially introduced by the surrogate models themselves in a cascaded prediction. For such concatenated integrated process models, the errors caused by the module predictions accumulate progressively through the sequence. This error accumulation can lead to a severe deviation in the final model's prediction results ($\hat{X}_K$) compared to the ground truth (X_K).

Therefore, further joint fine-tuning based on the pre-trained modular process models is necessary. This step adjusts the parameters of all cascaded modules simultaneously to minimize the accumulated error, particularly focusing on the accuracy of the final predicted state. This could be regarded as a multi-task learning approach, ensuring that the overall integrated model's parameters are optimized to serve the final prediction outcomes.

The fine-tuning process for the integrated model is executed over E iterations to optimize the parameters θ of the K -step MPNet modules. Initially, the model is loaded with pre-trained MPNet weights. During each training epoch, given an initial state X_0 and a sequence of process conditions $[C_1, \dots, C_K]$, the model performs a sequential forward pass. Specifically, for each time step $k \in [1, \dots, K]$, the module M_k predicts the current state $\hat{X}_k$ using the predicted state from the previous step $\hat{X}_{k-1}$ and the corresponding condition C_k . To evaluate the cumulative performance, a total loss $\mathcal{L}$ is computed as the weighted sum of individual step losses, defined as $\mathcal{L} = \sum \mathcal{L}_k(\hat{X}_k, X_k)$, where $\hat{X}_k$ represents the ground truth. Finally, the parameters θ across all K modules are updated simultaneously through back-propagation based on the gradient of the total loss.

We compute the loss for each individual module within the integrated process. Instead of a simple summation, we introduce a sensitivity-weighted loss function to address the varying impact of different process steps on the final device performance. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^K \lambda_k \cdot \mathcal{L}_k(\hat{X}_k, X_k). \quad (6)$$

Which is subsequently back-propagated through the entire cascaded framework—from the final module back to the first—to mitigate the accumulation of errors. The λ is learn-

able and used to assign different weights to different process steps.

Based on the above joint training strategy, multi-task learning for MPNet can be realized, allowing the modular process model to not only learn the loss of the current process step but also learn from the loss propagated back from subsequent process steps. Following the dual-phase training strategy (pre-training of modules and joint fine-tuning of integrated process model), the MPNet model achieves step-by-step prediction of the integrated process. Compared to end-to-end models, this modular approach provides greater explainability. Simultaneously, the error accumulation resulting from the cascading of multiple process steps is effectively suppressed by the joint fine-tuning phase.

2.5. Dataset preparing methods

To train the modular surrogate model, a comprehensive dataset is required, generated from numerical process TCAD simulations. The generation process consists of two main stages: recipe sampling using DOE (design of experiments) and data pre-processing.

Given the high dimensionality of the process recipes ranges for an entire fabrication flow, an efficient sampling strategy is required to ensure the input variables which are evenly distributed across the design space. The OLHS (optimal latin hypercube sampling) method, which is superior to standard random sampling for high-dimensional spaces, is adopted to ensure better coverage and uniformity.

The maximum minimum distance algorithm is used to optimize the Latin Hypercube sampling. This algorithm aims to maximize the minimum distance between any two sampling points. Let $l(x_i, x_j)$ represent the distance between any two sampling points x_i and x_j in the d -dimensional design space. For a dataset of p experiments (sampling points), the objective is to maximize:

$$\min_{1 \leq i, j \leq p, i \neq j} l(x_i, x_j). \quad (7)$$

The distance $l(x_i, x_j)$ is typically the Euclidean ($t = 2$) or Manhattan ($t = 1$) distance:

$$l(x_i, x_j) = l_{ij} = \left[\sum_{k=1}^d |x_{ik} - x_{jk}|^t \right]^{1/t}. \quad (8)$$

Thus, the recipe parameters covering the whole process condition range can be obtained, and their uniformity can ensure that the test dataset is still evenly distributed in the experimental space.

After data generation, the final integrated process recipes are obtained. For each set of sampled recipes, the corresponding process flow gradually generates process results in simulation. Structure processes information is exported as index image matrices M_{struct} with each index representing a material.

Crucially, since the doping concentration in semiconductor devices typically spans several orders of magnitude, a direct linear mapping to pixel intensity would result in significant information loss, particularly for low-concentration regions. To address this, a logarithmic transformation is applied to the doping profile before normalization:

$$l(x, y) = \frac{\log_{10}(D(x, y)) - \log_{10}(D_{\min})}{\log_{10}(D_{\max}) - \log_{10}(D_{\min})}. \quad (9)$$

Here $D(x, y)$ is the local doping concentration, and $l(x, y)$ is the normalized grayscale intensity. The doping profile is thus exported as grayscale intensity maps M_{doping} with gray levels linearly corresponding to the logarithmic scale of dopant density. This logarithmic mapping ensures that the variations across the entire dynamic range are accurately captured and preserved within the floating-point precision of the image format, minimizing accuracy loss during the rasterization process.

2.6. Scalability to 3D architectures

While this study demonstrates the MPNet framework using a 2D planar MOSFET process, its modular design is inherently extensible to advanced 3D architectures such as FinFET and GAAFET. Extending MPNet to 3D scenarios primarily involves upgrading the core UNet-Attention modules to 3D-UNet architectures, utilizing 3D convolution kernels to process voxelized geometric and doping profiles.

To address concerns regarding model scale and computational complexity, we provide a quantitative estimation based on the architectural scaling laws. Regarding parameter size, extending convolution kernels from 2D ($k \times k$) to 3D ($k \times k \times k$) typically scales the parameters per layer by a factor of k (where $k = 3$). Given our current 2D module size of approximately 1.7 M parameters, a corresponding 3D module is estimated to contain between 5.5 and 7 M parameters. For an integrated 18-step process flow, by leveraging the proposed parameter sharing mechanism, the total parameter count of a 3D MPNet framework is projected to remain within 25 M. This scale fits comfortably within the memory constraints of modern high-end GPUs, ensuring the feasibility of large-scale parallel inference without the need for massive computing clusters.

In terms of computational complexity, assuming a 3D voxel resolution of $128 \times 128 \times 128$ (≈ 2.1 M voxels) compared to the current 512×256 pixel resolution (approx 0.13 M pixels), the data scale increases by approximately 16 $\times$. Combined with the computational overhead of 3D convolutions ($\approx 3\times$), the inference FLOPs are estimated to increase by a factor of ~ 50 . However, even if the single-step inference time increases from the current millisecond range (~ 0.01 s) to the sub-second range (~ 0.5 s), the advantage over numerical TCAD becomes even more pronounced. Since 3D TCAD simulation time scales super-linearly with the increase in mesh complexity, often incurring substantial time costs per split, the efficiency improvement provided by MPNet in the 3D field will still remain significant. Consequently, the speedup ratio is projected to exceed $10^3 \times$.

The current implementation primarily serves to validate the methodological foundation of the modular surrogate framework using a mature planar process. Based on the quantitative scalability analysis presented above, extending this methodology to 3D simulations is theoretically and computationally viable. This feasibility provides a solid basis for applying MPNet to advanced technology nodes, such as FinFET and GAA. In these regimes, where device performance is critically sensitive to 3D geometric variations (e.g., fin shape fidelity or nanosheet uniformity), traditional numerical TCAD

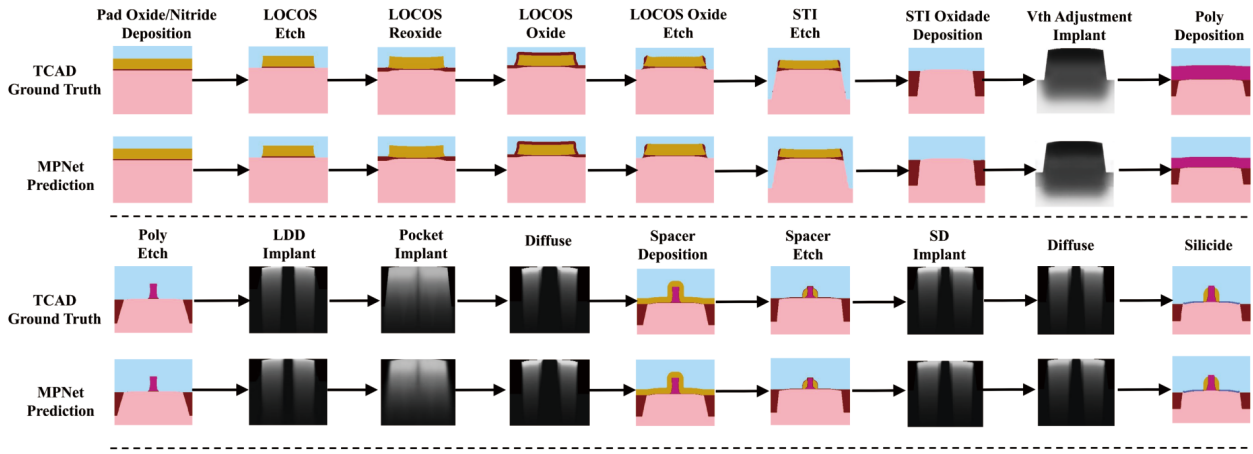


Fig. 4. (Color online) The step-by-step prediction of the integrated process.

faces severe computational bottlenecks. MPNet, with its demonstrated ability to capture complex process-structure dependencies, offers a promising solution for efficient process variation analysis and rapid DTCO (design technology co-optimization) in next-generation semiconductor manufacturing.

3. Experiments

This Section presents the experimental evaluation of the MPNet framework. It first introduces the MOSFET validation case study. Subsequently, the benchmark datasets, baseline methods, and evaluation metrics are described. The detailed results for both the modular and integrated models are then presented, followed by ablation studies to evaluate key components.

To validate the proposed framework, we consider a representative planar MOSFET process flow as a case study. Although process technology nodes have evolved toward more advanced device architectures, the main process components remain largely universal. The chosen planar CMOS nodes (e.g., 180 and 90 nm) are widely deployed in automotive electronics and power-management ICs, and provide a clean, well-characterized manufacturing environment in which these mechanisms can be isolated and analyzed. Their process steps are mature and well understood, with standardized flows and stable device characteristics. At the same time, the flow is sufficiently complex to capture the challenges of industrial fabrication, including multi-step film deposition, etching, ion implantation, and thermal diffusion sequences that mirror modern processes. This balance of maturity and complexity makes the chosen process an ideal testbed for evaluating the surrogate model's ability to learn cumulative process effects while remaining computationally tractable for generating a comprehensive training dataset.

3.1. Planar MOSFET integrated process flow

As shown in the later experiment results in Fig. 4, the full integrated process flow under consideration comprises 18 sequential process steps. The process initiates with substrate preparation, followed by the formation of isolation structures through LOCOS (local oxidation of silicon) and STI (shallow trench isolation) steps. Subsequently, the channel is engineered via a V_{th} adjustment implant. The polysilicon gate is then deposited and etched to define the gate structure. The

flow proceeds with comprehensive source/drain engineering, which includes LDD (lightly doped drain) implantation, pocket implantation, spacer deposition and etching, and the main S/D implantation, interspersed with critical diffusion and anneal steps to activate dopants. The integrated process sequence concludes with the formation of silicide to reduce contact resistance.

3.2. Experimental setup

Model and training parameters. As detailed in Subsection 2.3, each module (e.g., etch, deposition, implant) contains approximately 1.7 M parameters. The modules were trained using the Adam optimizer. For the pre-training phase, the learning rate was set to 1×10^{-4} with a batch size of 32 for 100 epochs. For the joint fine-tuning phase, the learning rate was reduced to 1×10^{-5} for an additional 50 epochs. All experiments were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU.

Evaluation metrics. To quantitatively evaluate the surrogate models, we employ metrics tailored to two prediction tasks: device geometry prediction (etch/film) and doping profile prediction (implant/diffuse).

- **IOU (intersection over union)**^[20]: For the etch and film process modules, which output device structure segmentations, we use IOU (jaccard index) to measure the spatial overlap between the predicted segmentation mask A and the ground-truth mask B .

$$\text{IOU}(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \quad (10)$$

Here TP, FP, and FN denote the numbers of true-positive, false-positive, and false-negative pixels, respectively.

- **MSE (mean squared error) and PSNR (peak signal-to-noise ratio)**^[21]: To evaluate the accuracy of predicted doping profiles (grayscale images), we adopt MSE and PSNR. MSE quantifies the average pixel-wise squared difference between the predicted image I and the ground-truth image K :

$$\text{MSE} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (I(i, j) - K(i, j))^2. \quad (11)$$

Here H and W represent the height and width of the image in pixels, respectively, and (i, j) denotes the pixel coordinate.

Table 1. Selected recipes and the irranges for the integrated MOSFET process case study.

Process Parameters	Max	Min
Lgate (μm)	0.188	0.172
Slice angle ($^\circ$)	42.0	48.0
Init boron (cm^{-2})	9.0×10^{14}	1.2×10^{15}
Diffuse1 temp ($^\circ\text{C}$)	990	1010
PadO _x thickness ($\text{\AA}$)	300	700
Drivein temp ($^\circ\text{C}$)	970	1030
PadNT (s)	0.1	0.3
LOCOS mask thickness (μm)	0.2	0.8
LOCOS etch time (s)	1.6	2.0
LOCOS oxide temp ($^\circ\text{C}$)	880	920
TEOS ($\text{\AA}$)	12	28
STI depth ($\text{\AA}$)	0.3	0.7
STI tilt ($^\circ$)	70	90
STI rate ($\text{\AA}/\text{s}$)	0.013	0.017
STI time (s)	1.95	2.0
Vth dose (cm^{-2})	1.00×10^{12}	7.00×10^{12}
Vth energy (keV)	0.7	1.3
Diffuse2 time (s)	0.05	0.07
Divot ($^\circ$)	33	93
Tpoly (s)	0.2	0.4
POLY etch (s)	0.35	0.39
PolyReox thickness (μm)	0.005	0.015
LDD energy (keV)	13	23
LDD dose (cm^{-2})	8.00×10^{14}	2.00×10^{15}
Pocket energy (keV)	22	28
Pocket dose (cm^{-2})	8.00×10^{12}	2.00×10^{13}
Spacer etch time (s)	0.42	0.46
NatOxide (μm)	0.005	0.007
S/D Ar energy (keV)	30	70
S/D P energy (keV)	70	90
TiSiT (s)	0.01	0.05

PSNR is derived from MSE to measure the ratio between the maximum possible signal power and the power of reconstruction error:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_i^2}{\text{MSE}} \right). \quad (12)$$

Here MAX_i is the maximum possible pixel value of the image (e.g., 255 for 8-bit images).

• **SSIM (structural similarity index)**^[22]: As a generic metric applied to all process modules, SSIM is used to evaluate the perceptual similarity between the model output and the ground truth. It assesses fidelity by comparing three components: luminance, contrast, and structure:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}. \quad (13)$$

Here x and y represent the sliding windows of the predicted and ground-truth images, respectively. μ_x and μ_y are the local means, σ_x^2 and σ_y^2 are the local variances, and σ_{xy} represents the local covariance between x and y . The terms $c_1 = (k_1L)^2$ and $c_2 = (k_2L)^2$ are constants to stabilize the division with weak denominator, where L is the dynamic range of pixel values.

Table 2. MPNet process modules results (%).

Step	Description	SSIM	IOU	PSNR
1	PAD deposition	99.6	99.9	–
2	LOCOS etch	99.6	99.8	–
3	LOCOS reoxide	99.4	99.2	–
4	LOCOS oxide	98.8	99.6	–
5	LOCOS oxide etch	99.0	99.2	–
6	STI etch	98.2	98.5	–
7	STI oxide deposition	98.9	99.7	–
8	V_{th} adjustment implant	97.5	–	34.8
9	Poly deposition	99.8	99.9	–
10	Poly etch	99.6	99.7	–
11	LDD implant	97.8	–	36.7
12	Pocket implant	97.2	–	35.1
13	LDD diffuse	97.4	–	49.9
14	Spacer deposition	99.2	99.7	–
15	Spacer etch	99.4	99.9	–
16	SD implant	98.6	–	40.2
17	SD diffuse	98.8	–	47.7
18	Silicide	99.1	99.6	–

3.3. Dataset preparation

We apply the dataset generation methodology described in Subsection 2.5 to an integrated MOSFET process flow. Although the proposed workflow is compatible with various open-source and commercial semiconductor TCAD tools, in this work we adopt Synopsys Sentaurus TCAD as a representative implementation^[23–25].

Following the DOE procedure in Subsection 2.5, we first identify the key process parameters that strongly influence the MOSFET structure. Leveraging semiconductor manufacturing experience for this technology node, we then select 31 critical process recipes and specify their allowable fluctuation ranges, as summarized in Table 1. These ranges are chosen such that the resulting device simulations exhibit electrical characteristics within a normal operating window, thereby ensuring the practical relevance and reliability of the generated dataset. Based on these ranges, we use the OLHS method to sample 2000 distinct process-recipe combinations for the integrated process-flow simulations.

For each sampled recipe, we run the complete Sentaurus TCAD process-flow simulation and record the intermediate states. At each relevant process step, the corresponding structural and doping-profile images are exported via the Sentaurus Visual module. The resulting images are subsequently processed using the pre-processing pipeline defined in Subsection 2.5 to obtain Mstruct and Mdoping with a resolution of 512×256 .

3.4. Process modules' results

For MPNet modules, we obtained predictions and TCAD ground truth for each step of the process, and evaluated the test results for each step, as shown in Table 2.

Our modular models achieved more than 0.99 SSIM and IOU on SSIM of the structural process steps and more than 0.97 SSIM and 30 PSNR on the implantation processes. It can be clearly observed that our modular process models can fit the morphology and internal doping changes of NMOS devices caused by each step of the process.

Meanwhile, to demonstrate the performance of the mod-

Table 3. Comparison of evaluation metrics (IOU/SSIM/MSE) under various processes (%).

Process	Method	IOU	SSIM	MSE
Film	RTT ^[18]	96.8	96.2	4.2
	Pix2pix ^[26]	97.5	98.1	2.8
	DeeplabV3+ ^[7]	97.2	97.9	3.1
	MPNet	99.1	99.7	1.2
Etch	RTT ^[18]	96.5	95.9	4.5
	Pix2pix ^[26]	97.8	98.2	2.6
	DeeplabV3+ ^[7]	97.4	98.0	2.9
	MPNet	99.2	99.4	1.3
Imp	RTT ^[18]	/	95.5	4.9
	Pix2pix ^[26]	/	97.5	3.2
	DeeplabV3+ ^[7]	/	97.2	3.5
	MPNet	/	98.8	1.7
Integrated	RTT ^[18]	95.6	95.1	4.8
	Pix2pix ^[26]	96.9	97.2	3.6
	DeeplabV3+ ^[7]	97.5	97.8	3.0
	MPNet	98.5	98.7	2.1

Table 4. Comparison of 50 integrated process runs predicting time (s).

Step	TCAD	MPNet
PAD deposition	52	0.0115
LOCOS etch	8	0.0036
LOCOS reoxide	40	0.0112
LOCOS oxide	21	0.0031
LOCOS oxide etch	47	0.0030
STI etch	11	0.0113
STI oxide deposition	57	0.0104
Vth adjustment implant	62	0.0553
Poly deposition	9	0.0223
Poly etch	17	0.0202
LDD implant	18	0.0358
Pocket implant	50	0.0432
LDD diffuse	103	0.0221
Spacer deposition	11	0.0058
Spacer etch	17	0.0056
SD implant	47	0.0233
SD diffuse	127	0.0228
Silicide	42	0.0032
Integrated	749	0.3134

ules, we conducted comparative experiments with RTT^[18], DeeplabV3+^[7], and the pix2pix model^[26], which are commonly used in similar tasks. Although the original authors did not provide specific model code, we reproduced the implementation based on their proposed architecture to conduct inference. We compare the accuracy of all process modules. As shown in Table 3, MPNet outperforms RTT, DeepLabV3+, and pix2pix, achieving average accuracy improvements of 0.8, 1.2, and 1.6 percentage points in 3 typical process modules, respectively.

3.5. Integrated process model's results

After following the method described in Subsection 2.4 to fine-tune the cascaded modules, we obtained a surrogate model for the whole 18-step integrated process. Fig. 4 provides a comprehensive qualitative validation of the integrated process MPNet, presenting a step-by-step visual com-

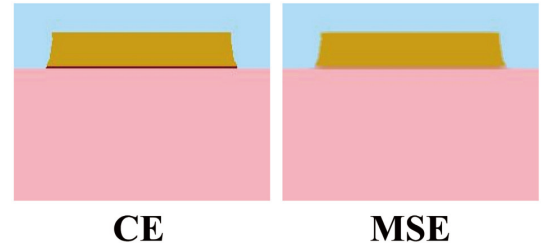


Fig. 5. (Color online) The performance of MPNet using MSE/CE as the loss function.

Table 5. Comparison of parameter quantity before and after parameter sharing (M).

Modules	Model Num	Before	After
Etch	5	8.75	1.76
Film	7	12.25	1.76
Implant	6	10.5	1.78

Table 6. MPNet's etch module SSIM accuracy before and after parameter sharing (%).

Models	Before	After
LOCOS-etch	99.6	99.6
LOCOS-OX-etch	99.0	98.8
STI-etch	98.2	97.9
Poly-etch	99.6	99.4
Spacer-etch	99.7	99.6

parison between the MPNet-predicted device states and the corresponding TCAD simulation ground truth. This sequential visualization demonstrates the MPNet's high fidelity in accurately capturing the complex evolution of device geometry and doping profiles across the entire process flow.

To compare the performance of different existing models on the integrated process, we also share the parameters of RTT, pix2pix, and DeeplabV3+ process modules, and calculate their accuracy in Table 3. The result shows that our MPNet outperforms other comparison methods in multiple metrics throughout the integrated process. This result underscores the robustness of the MPNet framework, highlighting the efficacy of its modular design and dual-phase joint fine-tuning strategy in mitigating error accumulation.

Finally, we compared the computation speed of MPNet and TCAD simulation for the process modules and integrated processes. Table 4 presents the average TCAD simulation time and the prediction time of our MPNet for process modules and integrated processes across 50 sets of integrated process experiments. In addition to this, benefiting from the parallel inference capability of GPUs, our MPNet can perform batch computation with a batch size (bs) of 16, and the time required is less than 16 times that when bs = 1. The results show that our MPNet demonstrates a nearly 2385 times speedup over TCAD.

3.6. Ablation study

Selection of loss function. As the input and output of structure processes, such as etch modules and film modules, are exported as index metrics M_{struct} , the loss function is determined as CE loss in Eq. (2). Meanwhile, the data for our task can also be exported as RGB images. We explored the predic-

Table 7. SSIM of ablation studies for different feature evolution methods (%)

Methods	Film	Etch	Implant	Full-loop
Early	98.4	98.2	96.2	97.2
Late	98.8	98.8	97.8	98.1
Concatenate	98.8	98.5	97.2	98.2
Dot product	99.2	99.1	97.6	98.6
MPNet	99.4	99.1	98.2	98.7

tive performance of MPNet using these two loss functions. As illustrated in Fig. 5, the inference results generated by the module trained with MSE loss are poor, especially on the fuzzy boundaries of different device regions, while the predictions generated by the module trained with CE metrics are significantly clearer.

Parameter sharing. To validate the efficacy of our parameter sharing strategy in Subsection 2.3, we compared the model's parameter count in Table 5 and predictive accuracy in Table 6 before and after sharing. As shown in Table 5, sharing reduced the total parameters for all 18 steps from 31.5 M to 5.3 M. Crucially, Table 6 shows that this significant 6x parameter reduction came with no significant loss in predictive accuracy (e.g., Poly-Etch SSIM 99.6% vs 99.4%), confirming the validity of our approach.

Feature evolution method. We calculated the accuracy of three process modules and the integrated process model via different process feature evolution methods. Table 7 shows that our MPNet's feature evolution position performs better than other typical methods. It also showed that the cross-attention structure used in our MPNet can more effectively reflect the output differences caused by feature evolution, especially in the implant model.

4. Application

In this Section, we demonstrate how MPNet can be used for integrated process optimization, namely, to identify process recipes that produce a device matching a predefined target structure. By replacing expensive TCAD simulations with fast surrogate evaluations, the $\sim 10^3\times$ acceleration provided by MPNet removes the traditional computational bottleneck in iterative, high-dimensional process tuning and makes it feasible to apply heuristic optimization algorithms to such inverse design problems.

4.1. Optimization strategy

The optimization problem refers to selecting a set of variables or parameters to achieve the optimal design objective while satisfying a series of relevant constraints. In this study, the objective is to find an optimal recipe vector x that minimizes the discrepancy between the final device structure predicted by our surrogate model, $\hat{P} = \text{MPNet}(x)$, and a desired target structure, P_{target} . The decision variables x consist of the D process parameters (e.g., $D = 30$ in our case study), which include key recipe values such as STI etch time, L_{gate} , V_{th} dose, and LDD energy. These variables are constrained within the valid process window defined during our dataset preparation (Subsection 2.5).

To solve this high-dimensional, non-linear optimization problem, we employ the PSO (particle swarm optimization) algorithm^[27], a population-based method well-suited for such

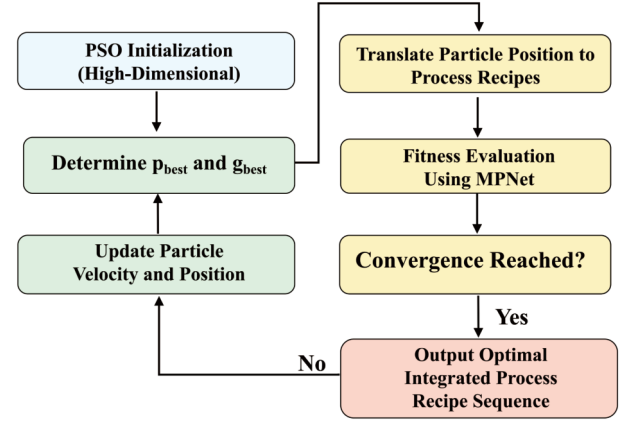


Fig. 6. (Color online) The PSO algorithm repeatedly calls MPNet as a fitness function to find the optimal recipes.

complex search spaces. The workflow is illustrated in Fig. 6. In this framework, each particle in the swarm represents a complete candidate recipe vector x_i . MPNet is critically invoked by the PSO algorithm at each iteration to evaluate the fitness of every particle. The algorithm uses this fitness score to update the particle's trajectory, guiding the swarm toward the global best solution (g_{best}).

Our objective is to minimize the discrepancy between the structure predicted by MPNet for a given recipe set (P_{pred}) and the corresponding ground truth (P_{true}). To jointly capture critical aspects of device geometry and doping, we define a composite fitness function. For a target process P_{true} and an MPNet-predicted process P_{pred} , the fitness is given by

$$\text{Fitness} = \alpha \cdot (1 - \text{SSIM}(P_{\text{true}}, P_{\text{pred}})) + \beta \cdot (1 - \text{IOU}(P_{\text{true}}, P_{\text{pred}})) + \gamma \cdot \text{MSE}(P_{\text{true}}, P_{\text{pred}}). \quad (14)$$

Here, SSIM, IOU, and MSE quantify the agreement between the predicted and reference processes. The terms $(1 - \text{SSIM})$ and $(1 - \text{IOU})$ convert the similarity indices into quantities to be minimized. The coefficients α , β , and γ are nonnegative weighting factors that satisfy $\alpha + \beta + \gamma = 1$ and are chosen empirically to balance structural integrity, regional overlap, and doping-profile accuracy. These weights can be tuned to emphasize specific device characteristics; for example, when optimizing a topography process (etch and film modules), we set $\gamma = 0$, whereas for a doping process (implant modules), we set $\beta = 0$. For full integrated-process optimization, all three terms are retained.

Process optimization is then performed using particle swarm optimization (PSO). The optimization loop is initialized by sampling a swarm of particles with random process recipes within the valid process corners. At each iteration, MPNet evaluates the fitness of every particle, and the swarm updates its personal-best (p_{best}) and global-best (g_{best}) positions, progressively converging toward an optimal recipe set. The final g_{best} vector is taken as the optimized recipe, which is subsequently validated against the ground-truth TCAD simulation to confirm its reliability.

4.2. Optimization results and discussion

To express the convergence curve of PSO like the convergence curve of model loss, we use the following formula:

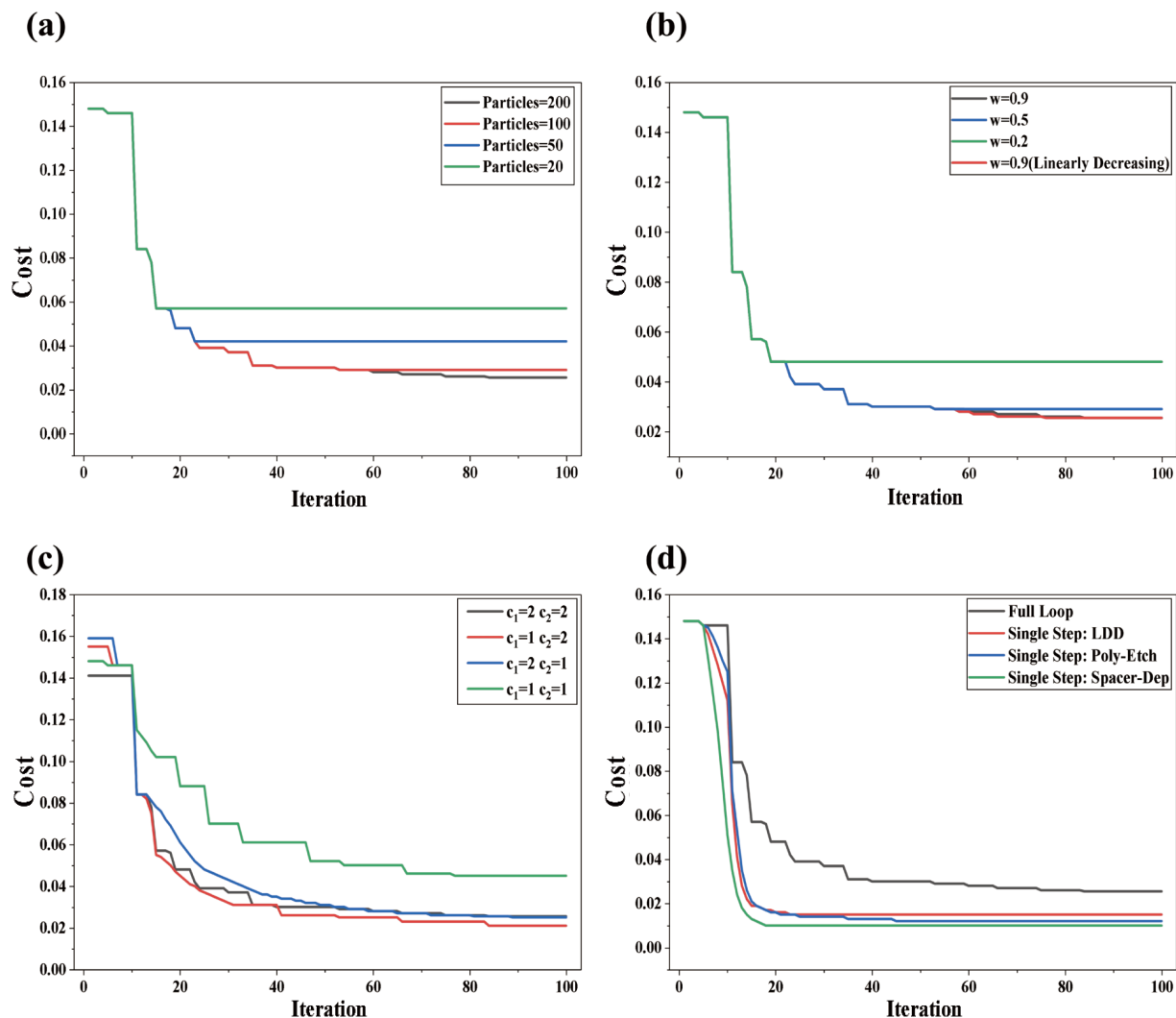


Fig. 7. (Color online) The cost decline curve of PSO. (a) Different number of particles, (b) different w , (c) different combinations of C_1 and C_2 , (d) integrated process and different single steps.

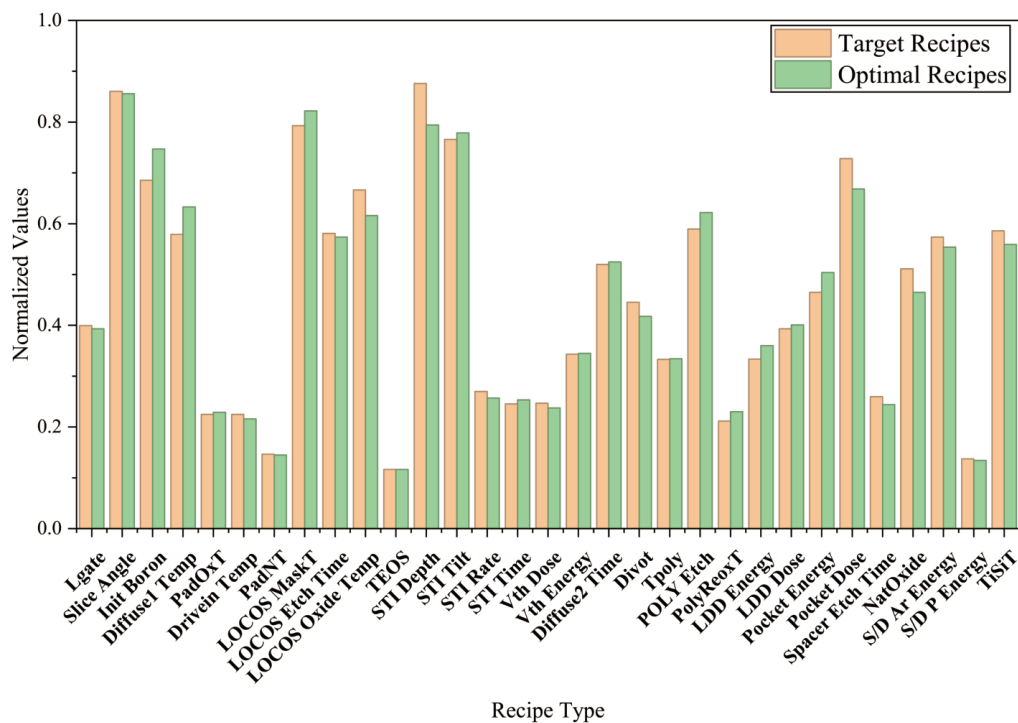


Fig. 8. (Color online) The distribution of the optimized recipe parameters normalized against the ground truth.

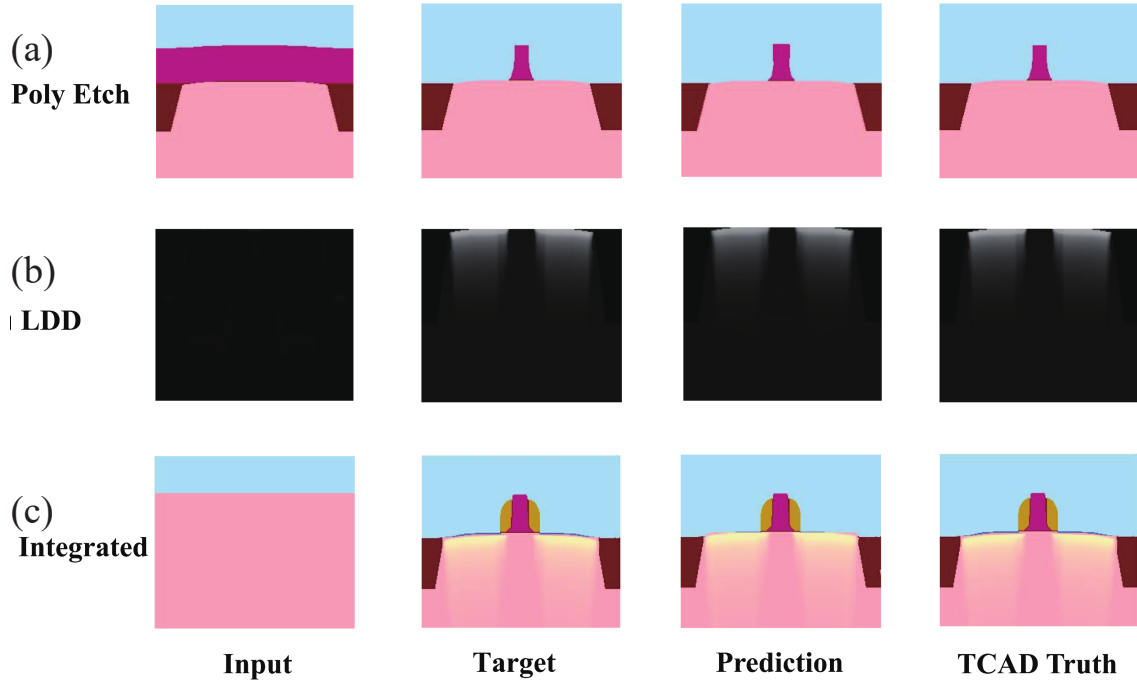


Fig. 9. (Color online) Visualization of target image, TCAD ground-truth image, and corresponding prediction using PSO optimal recipes. (a) Process module: Poly etch, (b) process module: LDD implant, (c) integrated process.

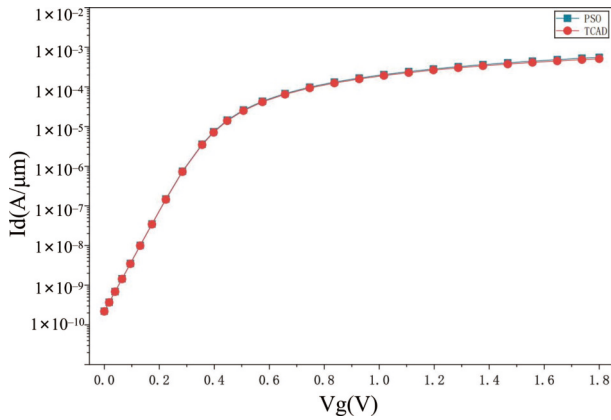


Fig. 10. (Color online) Comparison of the I - V test results of the devices obtained from the optimal recipe generated by PSO with the truth from TCAD.

$$\text{Cost} = 1 - \text{Fitness}. \quad (15)$$

Before applying the PSO algorithm to the target optimization problem, we first performed experimental tuning to determine the optimal hyperparameters, as detailed in Fig. 7. A swarm size of 100 was selected, as it provided the best trade-off between final cost and computational efficiency (Fig. 7(a)). The fastest convergence and best optimization outcome were achieved using a linearly decreasing inertia weight (w) combined with learning factors $c_1 = 1.5$ and $c_2 = 1.5$ (Figs. 7(b) and 7(c)). A final comparison confirmed that while the optimization for process modules converges faster to a lower cost, the algorithm also demonstrates effective convergence for the more complex, high-dimensional integrated process (Fig. 7(d)). All subsequent optimization experiments utilized these determined optimal parameters, converging to a cost of nearly 0.015.

With the tuned PSO algorithm, we executed the optimization strategy on both process modules and integrated pro-

cess targets. Fig. 8 displays a set of sampled recipes that were not present in our dataset, along with the normalized results of their corresponding optimal recipes obtained through MPNet optimization in process simulation. The results indicate that for the vast majority of recipes, MPNet can optimize to achieve values that are extremely close. Fig. 9 visualizes the final optimization results for three distinct cases: Poly etch, LDD implant, and the integrated process. For the process modules, the optimization target refers to a corresponding pair of input and output stages. For the integrated process, the input is always a silicon substrate, and only the final output image serves as the target to determine the optimal recipes. Subsequently, we used MPNet to predict the output states of the process modules and the integrated process via the obtained optimal recipes; meanwhile, we also input the sequence of optimal recipes into TCAD for re-simulation, obtaining both the process results and device electrical properties. The results in Fig. 9 and Fig. 10 show our PSO algorithm has demonstrated outstanding accuracy both in the optimization of low-dimensional process module recipes and the optimization of high-dimensional integrated process recipes, confirming MPNet's successful guidance of the PSO.

To illustrate the optimization process itself, Fig. 11 shows predictions of several structural steps in the integrated process before and after optimization. The output of these sequential process steps progressively approaches the target morphology as the algorithm iterates, demonstrating that MPNet effectively guides the high-dimensional search of the optimal recipes of the entire integrated process.

The primary benefit of the MPNet-PSO framework is the dramatic reduction in computational cost, enabling a task previously impractical with conventional TCAD. Taking the optimization process in Fig. 11 as an example, a single iteration using MPNet required only 5 min. In stark contrast, simulating the equivalent 100 recipe variations in TCAD would require over 60 000 min. For the 10 iterations required for con-

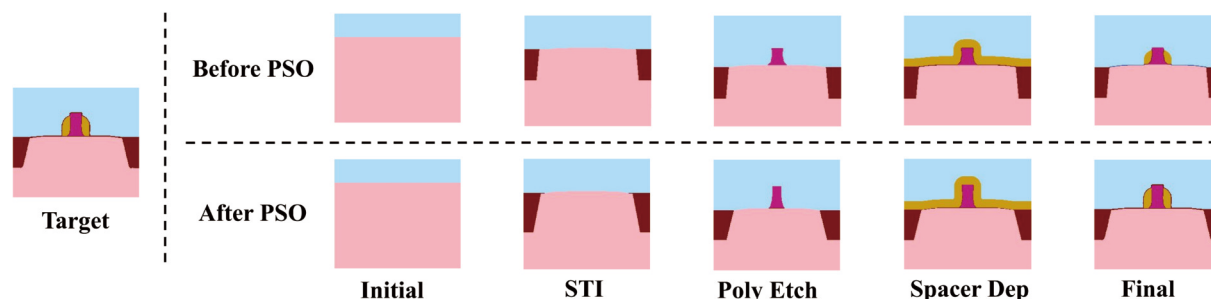


Fig. 11. (Color online) Visualization of the case MOSFET integrated process before & after PSO.

vergence, our framework achieves a time saving of over 99.9%.

In summary, this application demonstration confirms the practical utility of the MPNet framework. By coupling with the PSO algorithm, and thanks to its strong interpretable structure of multiple modules, MPNet enables the rapid, automated optimization of the high-dimensional process space—a task that remains computationally intractable for numerical TCAD simulations due to their prohibitive cost.

5. Conclusions

To address the prohibitive computational cost of TCAD and the limited generalizability of existing end-to-end surrogate models, this paper presented MPNet, a modular deep learning framework for process TCAD. MPNet decomposes complex integrated process flows into individual modules, each modeled by a UNet-Attention-based network to precisely capture the step-by-step evolution of device geometry and doping. A novel two-stage module pre-training and integrated fine-tuning strategy is employed to ensure high fidelity for individual process steps while effectively suppressing cumulative error in long cascades, achieving an overall integrated process fidelity of over 98.7% SSIM for an 18-step MOSFET flow. This approach yields a significant computational speedup of over 10^3 times compared to numerical TCAD. By coupling MPNet with a PSO algorithm, we demonstrated its practical utility in solving high-dimensional process recipes optimization problems, thereby providing a highly accurate, efficient, and explainable simulation paradigm that paves the way for accelerated IC development and tighter DTCO cycles.

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Qiwei Zhang is an Eng.D. student in the College of Integrated Circuits at Zhejiang University, beginning his studies in 2023. He received his B.S. degree in Microelectronics Science and Engineering from East China Normal University in 2023. His current research focuses on AI for Integrated Circuits Manufacturing.



Pengwei Liu is a Ph.D. student in the Department of Control Science and Engineering at Zhejiang University, beginning his studies in 2022. He received his B.S. degree in Mathematics and Applied Mathematics from Hefei University of Technology in 2022. His current research focuses on AI for Scientific Computing.



Wenzhang Fang is a Principal Investigator at the College of Integrated Circuits of Zhejiang University (ZJU) and a researcher at ZJU-Hangzhou Global Scientific and Technological Innovation Center (HIC-ZJU). His main research focuses on integrated circuit manufacturing processes, AI-based integrated circuit manufacturing, and 2D material integrated image sensor chips. He has published over 60 papers in high-level academic journals, which have been cited more than 2800 times.



Dong Ni is a Professor and Doctoral Advisor at the College of Integrated Circuits, Zhejiang University. He received his Ph.D. from the University of California, Los Angeles (UCLA) in 2005. His primary research interests focus on the application of multi-scale systems and artificial intelligence methods to smart manufacturing in advanced fields, particularly integrated circuits. He has published numerous high-impact papers at top-tier AI conferences, including those selected for spotlight presentations.



Yuting Kong is a ZJU100 Young Professor at the College of Integrated Circuits, Zhejiang University. She got her Ph.D. degree from the College of Control Science and Engineering, Zhejiang University in 2021. Her research interests focus on intelligent manufacturing of integrated circuits and automated design of analog circuits.